

O0001 (TITAN A16 - HAAS NGC FORMAT)
(PART: TITAN A16 v2.1)
(MATERIAL: Ta-Ti-Au-C MONOLITHIC BILLET)
(DATUM: TOP CENTER FACE, BORE CENTERLINE)
(UNITS: METRIC)
(DATE: 2026-01-22)

(=====)

(TOOL LIST)

(=====)

(T1 - D25 FACE MILL - ROUGHING)
(T2 - D3.0 CARBIDE DRILL - BORE PILOT)
(T3 - D3.0 REAMER - BORE FINISH)
(T4 - D6 FLAT END MILL - POCKET ROUGHING)
(T5 - D2 PCD BALL MILL - HELIX & FINISH)
(T6 - D2 DRILL - NODE POCKETS)

(=====)

(WORK COORDINATE SYSTEM)

(=====)

G54 (WCS - CENTER TOP FACE)
G17 (XY PLANE)
G21 (METRIC)
G40 G49 G80 (CANCEL MODES)
G90 (ABSOLUTE POSITIONING)

(=====)

(OPERATION 1: FACE TOP SURFACE)

(=====)

T1 M06
S1200 M03 (1200 RPM FACE MILL)
G00 X0 Y0
G43 H01 Z50.0 (TOOL LENGTH COMP)
M08 (COOLANT ON)

G00 Z2.0
G01 Z-0.2 F100 (FACE 0.2MM)
G01 X-17.0 F400
G03 X17.0 Y0 I17.0 J0 (SPIRAL FACE)
G00 Z50.0
M09 (COOLANT OFF)

(=====)

(OPERATION 2: DRILL CENTRAL BORE)
(=====)

T2 M06
S2500 M03
G00 X0 Y0
G43 H02 Z50.0
M08

G98 G81 X0 Y0 Z-10.5 R2.0 F80 (PECK DRILL)
G80
G00 Z50.0
M09

(=====)
(OPERATION 3: REAM BORE TO Ø3.03MM)
(=====)

T3 M06
S800 M03
G00 X0 Y0
G43 H03 Z50.0
M08

G01 Z2.0 F50
G01 Z-10.5 F25 (SLOW REAM FEED)
G01 Z2.0 F50
G00 Z50.0
M09

(=====)
(OPERATION 4: MILL TOP NODE POCKETS)
(=====)

T6 M06
S3000 M03
G00 Z50.0
G43 H06
M08

(NODE COORDINATES - 12 POSITIONS)
#500 = 5.0 (RADIUS)
#501 = 2.0 (POCKET DIAMETER)
#502 = -1.0 (POCKET DEPTH)

(NODE 1 - 0° - TI)
G00 X5.000 Y0.000

G98 G81 Z#502 R2.0 F100
G80

(NODE 2 - 30° - AU)
G00 X4.330 Y2.500
G98 G81 Z#502 R2.0 F100
G80

(NODE 3 - 60° - TI)
G00 X2.500 Y4.330
G98 G81 Z#502 R2.0 F100
G80

(NODE 4 - 90° - AU)
G00 X0.000 Y5.000
G98 G81 Z#502 R2.0 F100
G80

(NODE 5 - 120° - TI)
G00 X-2.500 Y4.330
G98 G81 Z#502 R2.0 F100
G80

(NODE 6 - 150° - AU)
G00 X-4.330 Y2.500
G98 G81 Z#502 R2.0 F100
G80

(NODE 7 - 180° - TI)
G00 X-5.000 Y0.000
G98 G81 Z#502 R2.0 F100
G80

(NODE 8 - 210° - AU)
G00 X-4.330 Y-2.500
G98 G81 Z#502 R2.0 F100
G80

(NODE 9 - 240° - TI)
G00 X-2.500 Y-4.330
G98 G81 Z#502 R2.0 F100
G80

(NODE 10 - 270° - AU)

G00 X0.000 Y-5.000
G98 G81 Z#502 R2.0 F100
G80

(NODE 11 - 300° - TI)
G00 X2.500 Y-4.330
G98 G81 Z#502 R2.0 F100
G80

(NODE 12 - 330° - AU)
G00 X4.330 Y-2.500
G98 G81 Z#502 R2.0 F100
G80

G00 Z50.0
M09

(=====)
(OPERATION 5: TRIPLE HELIX THREADS)
(=====)
T5 M06 (D2 PCD BALL MILL)
S8000 M03
G43 H05 Z50.0
M08

(HELIX PARAMETERS)
#510 = 12.5 (HELIX RADIUS - MM)
#511 = 3.333 (PITCH PER REVOLUTION - MM)
#512 = 3 (NUMBER OF STARTS)
#513 = 0 (CURRENT ANGLE)
#514 = 0 (Z POSITION)
#515 = 0 (X CALC)
#516 = 0 (Y CALC)

(APPROACH FIRST START POINT)
#515 = #510
#516 = 0
G00 X#515 Y#516
G00 Z2.0

(HELIX START 1 - 0° OFFSET)
WHILE [#513 LE 1080] DO1
 #514 = [#513 / 360.0] * [#511 / #512] - 10.0
 #515 = COS[#513] * #510

#516 = SIN[#513] * #510

IF [#513 EQ 0] THEN G01 Z#514 F100
G01 X#515 Y#516 Z#514 F250

#513 = #513 + 5.0
END1

G00 Z50.0

(HELIX START 2 - 120° OFFSET)

#513 = 120.0
#515 = COS[#513] * #510
#516 = SIN[#513] * #510
G00 X#515 Y#516
G00 Z2.0

WHILE [#513 LE 1200] DO2
#514 = [[#513 - 120.0] / 360.0] * [#511 / #512] - 10.0
#515 = COS[#513] * #510
#516 = SIN[#513] * #510

IF [#513 EQ 120.0] THEN G01 Z#514 F100
G01 X#515 Y#516 Z#514 F250

#513 = #513 + 5.0
END2

G00 Z50.0

(HELIX START 3 - 240° OFFSET)

#513 = 240.0
#515 = COS[#513] * #510
#516 = SIN[#513] * #510
G00 X#515 Y#516
G00 Z2.0

WHILE [#513 LE 1320] DO3
#514 = [[#513 - 240.0] / 360.0] * [#511 / #512] - 10.0
#515 = COS[#513] * #510
#516 = SIN[#513] * #510

IF [#513 EQ 240.0] THEN G01 Z#514 F100
G01 X#515 Y#516 Z#514 F250

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#513 = #513 + 5.0  
END3
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G00 Z50.0  
M09
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(=====)  
(  OPERATION 6: FINAL SURFACE FINISH  )  
(=====)  
T5 M06  
S10000 M03  
G43 H05 Z50.0  
M08
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(SPIRAL FINISH PASS ON OD)  
G00 X15.0 Y0 Z2.0  
G01 Z0 F50
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```
#520 = 0 (ANGLE)  
#521 = 15.0 (RADIUS)
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```
WHILE [#520 LE 1080] DO4  
  #515 = COS[#520] * #521  
  #516 = SIN[#520] * #521  
  #514 = [#520 / 360.0] * -0.5 - 10.0
```

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  G01 X#515 Y#516 Z#514 F300  
  #520 = #520 + 2.0  
END4
```

```
G00 Z50.0  
M09  
M05  
M30  
%
```

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(=====)  
(  MAZAK EIA/ISO FORMAT VERSION      )  
(=====)  
(CONVERT FANUC MACRO VARIABLES TO MAZAK)  
(#500-#520 → #1-#20 IN MAZAK)  
(G98/G99 → USE G98 ONLY)  
(M08/M09 → VERIFY COOLANT CODES)
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(G43 → G43 H__ Z__ IN MAZAK STYLE)